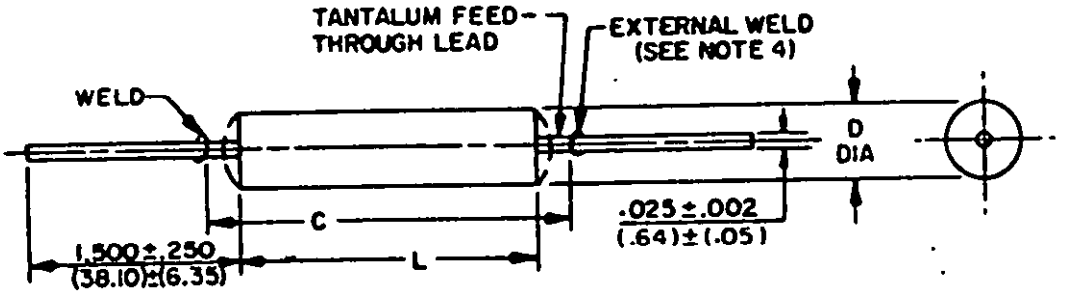
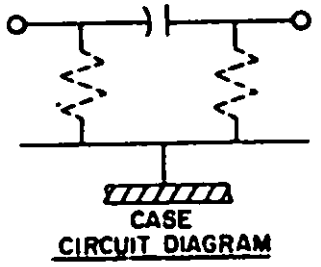
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (NONPOLARIZED, HIGH ETCHED FOIL), 85° C (VOLTAGE DERATED TO 125° C), ESTABLISHED RELIABILITY, STYLE CLR73

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39006.





Ⓐ

Case size	Dimensions			
	Basic case		Insulated case	
	L 1/ ±.062 (1.57)	D ±.016 (.41)	D Max	C Max
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
E2	.906 (23.01)	.281 (7.14)	.312 (7.92)	1.406 (35.71)
G2	.969 (24.61)	.281 (7.14)	.312 (7.92)	1.406 (35.71)
E3, G3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
E4, G4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
E5, G5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ Length of basic case; sleeving shall be as specified in MIL-C-39006.

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR73 capacitors.

Ⓐ denotes changes

FSC 5910

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TABLE I. Capacitor characteristics.

Rated voltage (85° C)	Cap.	Cap. tolerance	DC leakage (max)			Derated voltage (125° C)	Surge voltage (85° C)	Impedance (max)	Case size	Part No. M39006/17- Failure rate level (%/1,000 hr)				
			25° C	85° C	125° C					L(2.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
volts, dc	µF	%	µA	µA	µA	volts, dc	volts, dc	ohms						
15	20.0	+75, -15	5	24	16	10	17.2	185	E1	1000	1035	1070	1105	1140
15	70.0	+75, -15	15	84	56	10	17.2	55	E2	1001	1036	1071	1106	1141
15	200.0	+75, -15	40	240	160	10	17.2	18.5	E3	1002	1037	1072	1107	1142
15	410.0	+75, -15	75	500	330	10	17.2	8.9	E4	1003	1038	1073	1108	1143
15	590.0	+75, -15	118	710	470	10	17.2	6.2	E5	1004	1039	1074	1109	1144
25	10.0	+75, -15	4	20	12	15	28.8	370	E1	1005	1040	1075	1110	1145
25	35.0	+75, -15	12	70	43	15	28.8	106	E2	1006	1041	1076	1111	1146
25	100.0	+75, -15	30	200	120	15	28.8	37	E3	1007	1042	1077	1112	1147
25	210.0	+75, -15	40	400	250	15	28.8	17.5	E4	1008	1043	1078	1113	1148
25	360.0	+75, -15	100	600	360	15	28.8	12.1	E5	1009	1044	1079	1114	1149
30	8.0	+75, -15	3	19	12	20	34.5	456	E1	1010	1045	1080	1115	1150
30	29.0	+75, -15	11	60	40	20	34.5	137	E2	1011	1046	1081	1116	1151
30	80.0	+75, -15	30	180	120	20	34.5	45.6	E3	1012	1047	1082	1117	1152
30	170.0	+75, -15	65	390	260	20	34.5	22	E4	1013	1048	1083	1118	1153
30	240.0	+75, -15	90	550	370	20	34.5	7.6	E5	1014	1049	1084	1119	1154
50	3.5	+50, -15	3	14	9	30	57.5	1080	E1	1015	1050	1085	1120	1155
50	13.0	+50, -15	10	52	31	30	57.5	285	E2	1016	1051	1086	1121	1156
50	36.0	+50, -15	24	145	86	30	57.5	104	E3	1017	1052	1087	1122	1157
50	75.0	+50, -15	45	300	180	30	57.5	49	E4	1018	1053	1088	1123	1158
50	110.0	+50, -15	75	440	260	30	57.5	34	E5	1019	1054	1089	1124	1159
75	2.0	+50, -15	2	12	8	50	86.2	1850	E1	1020	1055	1090	1125	1160
75	7.0	+50, -15	8	42	28	50	86.2	550	E2	1021	1056	1091	1126	1161
75	20.0	+50, -15	23	120	80	50	86.2	185	E3	1022	1057	1092	1127	1162
75	43.0	+50, -15	45	260	170	50	86.2	86	E4	1023	1058	1093	1128	1163
75	60.0	+50, -15	60	360	240	50	86.2	62	E5	1024	1059	1094	1129	1164
100	1.4	+30, -15	2	11	7	65	115.0	2700	E1	1025	1060	1095	1130	1165
100	5.0	+30, -15	7	38	25	65	115.0	750	E2	1026	1061	1096	1131	1166
100	14.0	+30, -15	15	110	68	65	115.0	268	E3	1027	1062	1097	1132	1167
100	29.0	+30, -15	35	220	145	65	115.0	137	E4	1028	1063	1098	1133	1168
100	41.0	+30, -15	50	320	210	65	115.0	91	E5	1029	1064	1099	1134	1169
150	0.75	+30, -15	2	9	6	100	172.0	5000	E1	1030	1065	1100	1135	1170
150	2.7	+30, -15	6	30	20	100	172.0	1380	E2	1031	1066	1101	1136	1171
150	7.5	+30, -15	15	84	56	100	172.0	490	E3	1032	1067	1102	1137	1172
150	15.0	+30, -15	30	180	120	100	172.0	250	E4	1033	1068	1103	1138	1173
150	22.0	+30, -15	45	270	175	100	172.0	170	E5	1034	1069	1104	1139	1174

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TABLE I. Capacitor characteristics. - Continued.

Rated voltage (65° C)	Cap.	Cap. tolerance	DC leakage (max)			Derated voltage (125° C)	Surge voltage (85° C)	Impedance (max)	Case size	Part No. M39006/17- Failure rate level (%/1,000 hr)				
			25° C	85° C	125° C					L(2.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
volts, dc	µF	%	µA	µA	µA	volts, dc	volts, dc	ohms						
15	70.0	+75, -15	15	84	56	10	17.2	55	G2	2001	2036	2071	2106	2141
15	200.0	+75, -15	40	240	160	10	17.2	18.5	G3	2002	2037	2072	2107	2142
15	410.0	+75, -15	75	500	330	10	17.2	8.9	G4	2003	2038	2073	2108	2143
15	590.0	+75, -15	118	710	470	10	17.2	6.2	G5	2004	2039	2074	2109	2144
25	35.0	+75, -15	12	70	43	15	28.8	106	G2	2006	2041	2076	2111	2146
25	100.0	+75, -15	30	200	120	15	28.8	37	G3	2007	2042	2077	2112	2147
25	210.0	+75, -15	40	400	250	15	28.8	17.5	G4	2008	2043	2078	2113	2148
25	300.0	+75, -15	100	600	360	15	28.8	12.1	G5	2009	2044	2079	2114	2149
30	29.0	+75, -15	11	60	40	20	34.5	137	G2	2011	2046	2081	2116	2151
30	80.0	+75, -15	30	180	120	20	34.5	45.5	G3	2012	2047	2082	2117	2152
30	170.0	+75, -15	65	390	260	20	34.5	22	G4	2013	2048	2083	2118	2153
30	240.0	+75, -14	90	550	370	20	34.5	7.8	G5	2014	2049	2084	2119	2154
50	13.0	+50, -15	10	52	31	30	57.5	285	G2	2016	2051	2086	2121	2156
50	36.0	+50, -15	24	145	86	30	57.5	104	G3	2017	2052	2087	2122	2157
50	75.0	+50, -15	45	300	180	30	57.5	49	G4	2018	2053	2088	2123	2158
50	110.0	+50, -15	75	440	260	30	57.5	34	G5	2019	2054	2089	2124	2159
75	7.0	+50, -15	8	42	28	50	86.2	550	G2	2021	2056	2091	2126	2161
75	20.0	+50, -15	23	120	80	50	86.2	185	G3	2022	2057	2092	2127	2162
75	43.0	+50, -15	45	260	170	50	86.2	86	G4	2023	2058	2093	2128	2163
75	60.0	+50, -15	60	360	240	50	86.2	62	G5	2024	2059	2094	2129	2164
100	5.0	+30, -15	7	38	25	65	115.0	750	G2	2026	2061	2096	2131	2166
100	14.0	+30, -15	15	110	68	65	115.0	268	G3	2027	2062	2097	2132	2167
100	29.0	+30, -15	35	220	145	65	115.0	137	G4	2028	2063	2098	2133	2168
100	41.0	+30, -15	50	320	210	65	115.0	91	G5	2029	2064	2099	2134	2169
150	2.7	+30, -15	6	30	20	100	172.0	1380	G2	2031	2066	2101	2136	2171
150	7.5	+30, -15	15	84	56	100	172.0	490	G3	2032	2067	2102	2137	2172
150	15.0	+30, -15	30	180	120	100	172.0	250	G4	2033	2068	2103	2138	2173
150	22.0	+30, -15	45	270	175	100	172.0	170	G5	2034	2069	2104	2139	2174

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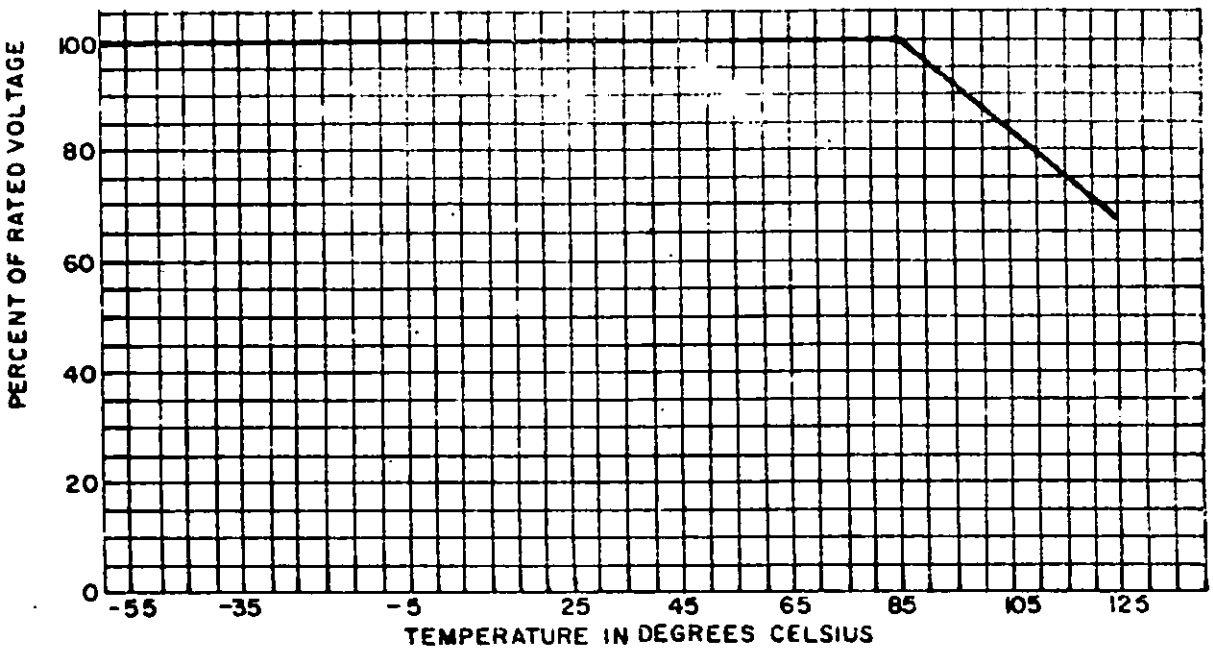


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular, insulated.
Ⓐ Seal type - G (hermetic) or E (nonhermetic) as shown in table I.
Terminals - Axial-wire lead.
Rated temperature - -55° to +85°C, voltage derated to +125°C (see figure 2).
Rated voltage - See table I.
Capacitance (Cap.) (nom) - See table I.
Cap. tolerance - See table I.
FR level - See table I.

Constant voltage conditioning: As specified in MIL-C-39006.

DC leakage (DCL) (max) (at 25°C): See table I.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max):
50 volts and below - 20%.
Above 50 volts - 15%.

Ⓐ Seal: (G seal only): Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).

Temperature cycling: Method 102 of MIL-STD-202, condition D (with step 3 at +125°C). Number of cycles - 30.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±5% of initial measurement.
DF - 115% (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hr).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:
Pull - Method 211 of MIL-STD-202, condition A.
Wire-lead bend - In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±6% of initial measurement.
DF - 115% (max) of initial requirement.

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Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202. 2,000 Vdc min.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100 megohms, min.

Low temperature (storage): Method 502 of MIL-STD-810.
- DCL - See table I.
- ΔCap. - Within ±5% of initial measurement.
- DF - 20% (≤50V) or 15% (>50V) (max).

Stability at low and high temperatures:
- Step 1 (+25°C):
  - DCL - See table I.
  - Cap. - Within tolerance of table I value.
  - DF - 20% (≤50V) or 15% (>50V) (max).
- Step 2 (-55°C):
  - Imp. - See table I.
  - ΔCap. - Within -40% of step 1 value.
- Step 3 (+25°C):
  - DCL - See table I.
  - ΔCap. - Within ±5% of step 1 value.
  - DF - 20% (≤50V) or 15% (>50V) (max).
- Step 4 (+85°C):
  - DCL - See table I.
  - ΔCap. - Within +50% of step 1 value.
  - DF - 55% (≤50V) or 45% (>50V) (max).
- 1/ Step 5 (+125°C):
  - DCL - See table I.
  - ΔCap. - Within +75% of step 1 value.
  - DF - 55% (≤30V) or 45% (>30V) (max).
- Step 6 (+25°C):
  - DCL - See table I.
  - ΔCap. - Within ±5% of step 1 value.
  - DF - 20% (≤50V) or 15% (>50V) (max).

Reverse voltage: Not applicable.

Life at 85°C: Method 108 of MIL-STD-202.
- 2,000-hr (qualification):
  - DCL - See table I.
  - ΔCap. - Within ±15% of initial measurement.
  - DF - 20% (≤50V) or 15% (>50V) (max).
- 240-hr (group B):
  - DCL - See table I.
  - ΔCap. - Within ±10% of initial measurement.
  - DF - 20% (≤50V) or 15% (>50V) (max).

10,000-hr (group C) and extended life:
- DCL (85°C) - 125% (max) of table I value.
- DCL (25°C) - See table I.
- ΔCap. - Within ±25% of initial measurement.
- DF - 40% (<50V) or 30% (>50V) (max).
- DWV - 2,000 Vdc, min.
- IR - 100 megohms, min.

Ⓐ Vacuum life: (G seal only):
- DCL - See table I.
- ΔCap. - Within ±20% of initial measurement.
- DF - 10% (max).
- Weight loss - 4.1 milligrams, max.

1/ Life at 125°C: Method 108 of MIL-STD-202.
- DCL - See table I.
- ΔCap - Within ±15% of initial measurement.
- DF - 20% (≤30V) or 15% (>30V) (max).
- DWV - 2,000 Vdc, min.
- IR - 100 megohms, min.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 ft).
- Test voltage - Rated voltage.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
- DCL - See table I.
- ΔCap. - Within ±5% of initial measurement.
- DF - 20% (≤50V) or 15% (>50V) (max).

Marking: In accordance with MIL-C-39006.

Voltage group (85°C rated voltage):

I - - - - - - - 15 to 150V incl

1/ 125°C derated voltage referenced.

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Navy - EC
Air Force - 11

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Navy - EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
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b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

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